MEDTRACE AI CLINICAL INTELLIGENCE SYSTEM

Feature Verification & Test Report

Environment: Node.js v22.13.0 • Express 4.19 • MongoDB Server • TypeScript 5.4
Scope: Registration, Login, Vitals, Appointments, Medical Reports, Nurses Active Checks
Execution: Automated Live Server Test Runner • October 6, 2026

✓ ALL 24 TESTS PASSED

OVERALL STATUS	FEATURES TESTED	EXECUTION TIME	SECURITY PROTOCOL
100% PASS	**6 / 6**	**1.18s**	**JWT + RBAC**
24 of 24 verified	Core clinical workflows	Real HTTP socket latency	Bearer token verified

Scope Statement: This report certifies automated validation of the 6 requested modules against the active MedTrace Express/MongoDB architecture. All test records were verified end-to-end and cleanly torn down.

1. User Registration Verification

4 / 4 PASSED

ID	TYPE	ENDPOINT	TEST CASE DESCRIPTION	TIME	STATUS
REG-01	POST	`/api/auth/register`	Register patient account with full credentials (name, email, password, phone, hospital, department, role='Patient').	276 ms	✓ PASS
REG-02	POST	`/api/auth/register`	Duplicate email prevention. Returns HTTP 400 with duplicate account rejection error message.	8 ms	✓ PASS
REG-03	POST	`/api/auth/register`	Input schema validation rejection (invalid email format, password under 8 characters, missing fields).	6 ms	✓ PASS
REG-04	POST	`/api/auth/register`	Provision Doctor, Nurse, and Head Nurse professional accounts for role-based downstream testing.	229 ms	✓ PASS

2. Authentication & Login Verification

4 / 4 PASSED

ID	TYPE	ENDPOINT	TEST CASE DESCRIPTION	TIME	STATUS
LOG-01	POST	`/api/auth/login`	Successful credential authentication, bcrypt password matching, JWT access and refresh token issuance.	69 ms	✓ PASS
LOG-02	POST	`/api/auth/login`	Invalid password rejection. Returns HTTP 401 Unauthorized with credential error.	63 ms	✓ PASS
LOG-03	POST	`/api/auth/login`	Non-existent user email rejection. Returns HTTP 401 Unauthorized.	6 ms	✓ PASS
LOG-04	GET	`/api/auth/me`	Access protected profile endpoint with Bearer token authentication header.	16 ms	✓ PASS

3. Patient Log Vitals Verification

4 / 4 PASSED

ID	TYPE	ENDPOINT	TEST CASE DESCRIPTION	TIME	STATUS
VIT-01	POST	`/api/vitals`	Record vitals: HR=74 bpm, SpO_2=98%, BP=118/78 mmHg, Temp=98.4°F, Source='manual'.	20 ms	✓ PASS
VIT-02	GET	`/api/vitals/:id/latest`	Fetch most recent vitals reading, confirming update order and value integrity (HR=85, SpO_2=99%).	19 ms	✓ PASS
VIT-03	GET	`/api/vitals/:id`	Retrieve complete longitudinal vitals timeline history for patient.	7 ms	✓ PASS
VIT-04	POST	`/api/vitals`	Security check: reject unauthenticated vitals logging request with HTTP 401.	1 ms	✓ PASS

Feature Verification & Test Report

Appointments, Medical Report Management & Nurses Duty Surveillance

✓ CONTINUED • 100% PASS

4. Doctor Appointment Verification

4 / 4 PASSED

ID	TYPE	ENDPOINT	TEST CASE DESCRIPTION	TIME	STATUS
APT-01	POST	`/api/appointments`	Book appointment with patientId, doctorId, future dateTime, reason, clinical notes, and 'pending' status.	18 ms	✓ PASS
APT-02	GET	`/api/appointments?patientId=`	Query appointment list filtered by patient identifier; verify patient appointment visibility.	12 ms	✓ PASS
APT-03	GET	`/api/appointments?doctorId=`	Query appointment list filtered by doctor identifier; verify clinical calendar visibility.	11 ms	✓ PASS
APT-04	PUT	`/api/appointments/:id/status`	Update appointment status lifecycle from 'pending' to 'confirmed'.	12 ms	✓ PASS

5. Medical Report Upload Verification

3 / 3 PASSED

ID	TYPE	ENDPOINT	TEST CASE DESCRIPTION	TIME	STATUS
REP-01	POST	`/api/test-results`	Multipart/form-data upload of diagnostic laboratory report file (CBC with Differential) with metadata.	21 ms	✓ PASS
REP-02	GET	`/api/test-results/:id`	Retrieve patient's diagnostic report list and verify uploaded file record presence.	8 ms	✓ PASS
REP-03	GET	`/api/test-results/file/:id`	Download and stream uploaded report file; verify payload byte integrity and biomarker content.	8 ms	✓ PASS

6. Nurses Active Check Verification

5 / 5 PASSED

ID	TYPE	ENDPOINT / COMPONENT	TEST CASE DESCRIPTION	TIME	STATUS
NUR-01	LOGIC	`dutyWindow(duty, now)`	Active Duty State Check: verifies that current timestamp falling within scheduled duty window returns state = 'active'.	< 1 ms	✓ PASS
NUR-02	LOGIC	`dutyWindow state cycle`	Comprehensive window transitions: future duty → 'scheduled', elapsed duty → 'completed', cancelled → 'cancelled'.	< 1 ms	✓ PASS
NUR-03	LOGIC	`assertDutyTransition()`	Attendance Lifecycle Transitions: not_checked_in → checked_in → on_break → checked_in → checked_out. Rejects invalid skips.	< 1 ms	✓ PASS
NUR-04	POST	`/api/nurse-assignments`	Assign Nurse to Patient in Ward ('Cardiology Ward 3B', shift='day'). Query assigned patient roster for nurse.	25 ms	✓ PASS
NUR-05	PUT	`/api/nurse-tasks/:id`	End-to-end task workflow: Head Nurse creates task → Head Nurse approves task → Nurse marks task completed.	30 ms	✓ PASS

Quality Assurance & Verification Certification

This validation run executed on a live instance of the MedTrace central server connected to MongoDB. All data created during the test lifecycle (test patients, mock clinical vitals, mock laboratory records, and duty assignments) was safely processed and cleanly torn down upon completion. The system meets all functional criteria for the evaluated modules.